MTH207 2023S2 Solution

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Donaldson Tan

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH207

Examination - July Semester 2023

Linear Algebra

Wednesday, 15 November 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Let $V = \left\{ \begin{bmatrix} a-b+c \\ 11b-c \\ 3a+5b \\ 7c \end{bmatrix} \middle| \, a, b, c \in \mathbb{R} \right\}$ be a set in $\mathbb{R}^4$.

Let $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix} \right\}$ be a set of three vectors in $\mathbb{R}^4$.

Question 1a

Determine whether S is a linearly independent set.

(3 marks)

Question 1b

Show that V is a subspace of $\mathbb{R}^4$ and state the dimension of V.

(6 marks)

Question 1c

Does $\begin{bmatrix} 1 \\ 10 \\ 8 \\ 4 \end{bmatrix}$ belong to V?

No

(3 marks)

Question 1d

Explain why $\left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 1 \\ 10 \\ 8 \\ 4 \end{bmatrix} \right\}$ forms a basis of $\mathbb{R}^4$.

(3 marks)

Question 2a

Compute a row echelon form of the following matrix.

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 2 & -2 & 1 & 6 \\ 5 & -4 & a & -9 \end{bmatrix}$$

(8 marks)

Question 2b

Hence, or otherwise, find the solution set for the following equations:

$$\begin{aligned} x - 2z &= 4 \\ 2x - 2y + z &= 6 \\ 5x - 4y + 5z &= -9 \end{aligned}$$

(6 marks)

Question 2c(i)

For which real numbers x do the vectors

$$\begin{bmatrix} x \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix}$$

not form a basis of $\mathbb{R}^3$.

(7 marks)

Question 2c(ii)

Let

$$W = \text{Span}\left\{ \begin{bmatrix} x \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} \right\}.$$

For each of the values of x that you find in Question 2c(i), what is the dimension of the subspace W?

(4 marks)

Question 3a

Calculate all possible values of k for which the matrix

$$A = \begin{bmatrix} 0 & 1 & k \\ 2 & k & -6 \\ 2 & 7 & 4 \end{bmatrix}$$

is invertible.

(8 marks)

Question 3b

If the matrix in Question 3a is invertible, and

$$\begin{bmatrix} 0 & 1 & k \\ 2 & k & -6 \\ 2 & 7 & 4 \end{bmatrix}^{-1} = \frac{1}{(14-2k)k-20} \begin{bmatrix} 4k+42 & 7k-4 & -k^2-6 \\ -20 & -2k & 2k \\ 14-2k & 2 & x \end{bmatrix}.$$

Find the value of x. -2

(6 marks)

Question 3c

Using the answers to obtain for Question 3a and Question 3b, solve the following system of liner equations:

$$\begin{aligned} y + 6z &= 1 \\ 2x + 6y - 6z &= 2 \\ 2x + 7y + 4z &= 1. \end{aligned}$$

(6 marks)

Question 4

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T\begin{bmatrix} x \\ y \\ \cancel{x} \end{bmatrix} = \begin{bmatrix} 5x - 2.5y + 10z \\ -x + 0.5y - 2z \end{bmatrix}.$$

z

Question 4a

Compute the matrix representation A of T with respect to the standard basis in domain and codomain.

(4 marks)

Question 4b

(i) Compute a basis for $\mathrm{Col}(A)$, the column space of A.

(ii) Find the dimension of $\mathrm{Nul}(A)$, the null-space of A.

(11 marks)

Question 4c

(i) Determine whether $\begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$ is in $\mathrm{Nul}(A)$.

(ii) Is $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ in $\mathrm{Col}(A)$?

(5 marks)

Question 5

Let A be the matrix

$$\begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}.$$

Denote vectors $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ by e_1 and $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ by e_2.

Question 5a

Show that e_1 and e_2 are eigenvectors of A and calculate the eigenvalues corresponding to e_1 and e_2.

(6 marks)

Question 5b

Determine a remaining eigenvector e_3 such that $\{e_1, e_2, e_3\}$ form a linearly independent set of eigenvectors of A.

(8 marks)

Question 5c

Solve for an invertible matrix P and a diagonal matrix D such that

$$P^{-1}AP = D.$$

(6 marks)

----- END OF PAPER -----

1(a)

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 11 & -1 & 0 \\ 3 & 5 & 0 & 0 \\ 0 & 0 & 7 & 0 \end{array}\right] \xrightarrow[r_3 = r_3 - 3r_1 - \frac{8}{11}r_2]{r_2 = \frac{1}{11}r_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 11 & -1 & 0 \\ 0 & 8 & -3 & 0 \\ 0 & 0 & 7 & 0 \end{array}\right]$$

$$\xrightarrow[r_4 = r_4 + \frac{7 \times 1}{25} r_3]{r_3 = -\frac{11}{25} r_3} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -1/11 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[r_1 = r_1 - r_3]{r_2 = r_2 + \frac{1}{11} r_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{r_1 = r_1 + r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore$ set S is linearly independent

(b)

$$v = \begin{bmatrix} a-b+c \\ 11b-c \\ 3a+5b \\ 7c \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix} + c\begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix}$$

let $v(a,b,c) = a\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix} + c\begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix}$

$$v(0,0,0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\alpha v(a_1,b_1,c_1) + \beta v(a_2,b_2,c_2) = \alpha\left(a_1\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix} + b_1\begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix} + c_1\begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix}\right) +$$

$$\beta\left(a_2\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix} + b_2\begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix}\right)$$

$$= (\alpha a_1 + \beta a_2)\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix} + (\alpha b_1 + \beta b_2)\begin{bmatrix} -1 \\ 11 \\ 5 \\ 0 \end{bmatrix} + (\alpha c_1 + \beta c_2)\begin{bmatrix} 1 \\ -1 \\ 0 \\ 7 \end{bmatrix})$$

$$= v(\alpha a_1 + \beta a_2,\ \alpha b_1 + \beta b_2,\ \alpha c_1 + \beta c_2)$$

$\therefore$ V is in the subspace of $\mathbb{R}^4$

1(c)

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 11 & -1 & 10 \\ 3 & 5 & 0 & 8 \\ 0 & 0 & 7 & 4 \end{array}\right] \xrightarrow[]{\substack{r_2 = \frac{1}{11} r_2 \\ r_3 = r_3 - 3r_1 - \frac{8}{11} r_2}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & 0 & -25/11 & -25/11 \\ 0 & 0 & 7 & 4 \end{array}\right]$$

$$\xrightarrow[]{\substack{r_3 = -\frac{11}{25} r_3 \\ r_4 = r_4 + \frac{7 \times 11}{25} r_3}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

$\therefore$ The linear system is inconsistent

$\therefore \begin{bmatrix} 1 \\ 10 \\ 8 \\ 4 \end{bmatrix}$ doesn't belong to V

(d)

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 11 & -1 & 10 \\ 3 & 5 & 0 & 8 \\ 0 & 0 & 7 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$\therefore$ There are 4 pivots

$\therefore$ The set is linearly independent

$\therefore$ The set forms a basis of $\mathbb{R}^4$

2(a)
$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 2 & -2 & 1 & 6 \\ 5 & -4 & a & -9 \end{bmatrix} \xrightarrow[r_3 = r_3 - 5r_1 - 2r_2]{r_2 = -\frac{1}{2}(r_2 - 2r_1)} \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & -5/2 & 1 \\ 0 & 0 & a & -25 \end{bmatrix}$$

$$\xrightarrow[\substack{r_2 = r_2 + \frac{5}{2a} r_3 \\ r_1 = r_1 + \frac{2}{a} r_3}]{r_3 = \frac{1}{a} r_3} \begin{bmatrix} 1 & 0 & 0 & 4 - \frac{50}{a} \\ 0 & 1 & 0 & 1 - \frac{125}{2a} \\ 0 & 0 & 1 & -\frac{25}{a} \end{bmatrix}$$

(b)
$$\begin{aligned} x - 2z &= 4 \\ 2x - 2y + z &= 6 \\ 5x - 4y + 5z &= -9 \end{aligned} \iff \begin{bmatrix} 1 & 0 & -2 & 4 \\ 2 & -2 & 1 & 6 \\ 5 & -4 & 5 & -9 \end{bmatrix} \quad \therefore a = 5$$

$\therefore x = 4 - \frac{50}{a} = 4 - 10 = -6$

$\therefore y = 1 - \frac{125}{2a} = -11\frac{1}{2}$

$\therefore z = -\frac{25}{a} = -5$

(c) i.
$$\left[\begin{array}{ccc|c} x & 1 & 1 & 0 \\ 1 & x & 1 & 0 \\ 1 & 1 & x & 0 \end{array}\right] \xrightarrow{\text{swap } r_1, r_3} \left[\begin{array}{ccc|c} 1 & 1 & x & 0 \\ 1 & x & 1 & 0 \\ x & 1 & 1 & 0 \end{array}\right] \xrightarrow[r_3 = r_3 - xr_1]{r_2 = r_2 - r_1} \left[\begin{array}{ccc|c} 1 & 1 & x & 0 \\ 0 & x-1 & 1-x & 0 \\ 0 & 1-x & 1-x^2 & 0 \end{array}\right]$$

$$\xrightarrow{r_3 = r_3 + r_2} \left[\begin{array}{ccc|c} 1 & 1 & x & 0 \\ 0 & x-1 & 1-x & 0 \\ 0 & 0 & -(x+2)(x-1) & 0 \end{array}\right]$$

To form a basis, there must be 3 pivots.

Hence, $x \neq 1$ or $x \neq -2$

$\therefore$ range of x that don't form a basis in $\mathbb{R}^3 = \{1, -2\}$

ii. when $x = 1$

$$\left[\begin{array}{ccc|c} x & 1 & 1 & 0 \\ 0 & 1-x^2 & 1-x & 0 \\ 0 & 0 & -x(x-1)(x+2) & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow 1 \text{ pivot}$$

$\therefore \dim W = 1$

when $x = -2$

$$\left[\begin{array}{ccc|c} x & 1 & 1 & 0 \\ 0 & 1-x^2 & 1-x & 0 \\ 0 & 0 & -x(x-1)(x+2) & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow 2 \text{ pivots}$$

$\therefore \dim W = 2$

3(a) let $\det A = 0$

$$\begin{vmatrix} 0 & 1 & k \\ 2 & k & -6 \\ 2 & 7 & 4 \end{vmatrix} = 0 \xrightarrow{r_3 = r_3 - r_2} \begin{vmatrix} 0 & 1 & k \\ 2 & k & -6 \\ 0 & 7-k & 10 \end{vmatrix} = 0$$

$$2\begin{vmatrix} 1 & k \\ 7-k & 10 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & k \\ 7-k & 10 \end{vmatrix} = 0$$

$$k^2 - 7k + 10 = 0$$

$$(k-5)(k-2) = 0$$

$\therefore A^{-1}$ doesn't exist when $k = 2$ or 5

$\therefore$ range of $k = \mathbb{R} \setminus \{2, 5\}$

(b) $A^{-1}A = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\Leftrightarrow \frac{1}{(14-2k)k-20}\begin{bmatrix} 14-2k & 2 & x \end{bmatrix}\begin{bmatrix} k \\ -6 \\ 4 \end{bmatrix} = 1$$

$$\begin{bmatrix} 14-2k & 2 & x \end{bmatrix}\begin{bmatrix} k \\ -6 \\ 4 \end{bmatrix} = (14-2k)k - 20$$

$$(14-2k)k - 12 + 4x = (14-2k)k - 20$$

$$4x = -8$$

$$\therefore x = -2$$

(c) $k = 6$

$$A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{8}\begin{bmatrix} 66 & 38 & -42 \\ -20 & -12 & 12 \\ 2 & 2 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{8}\begin{bmatrix} 100 \\ -32 \\ 4 \end{bmatrix}$$

4(a) $T(e_1) = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$

$T(e_2) = \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix}$

$T(e_3) = \begin{bmatrix} 10 \\ -2 \end{bmatrix}$

$\therefore A = \begin{bmatrix} 5 & -2.5 & 10 \\ -1 & 0.5 & -2 \end{bmatrix}$

(b) i. $\begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 5 \\ -1 \end{bmatrix}$

$\begin{bmatrix} 10 \\ -2 \end{bmatrix} = 2\begin{bmatrix} 5 \\ -1 \end{bmatrix}$

$\therefore$ basis for $\text{col}(A) = \left\{ \begin{bmatrix} 5 \\ -1 \end{bmatrix} \right\}$

ii. $\begin{bmatrix} 5 & -2.5 & 10 \\ -1 & 0.5 & -2 \end{bmatrix} \xrightarrow{r_2 = 5r_2 + r_1} \begin{bmatrix} 5 & -2.5 & 10 \\ 0 & 0 & 0 \end{bmatrix}$

$\downarrow r_1 = \frac{1}{5}r_1$

$\begin{bmatrix} 1 & -0.5 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore \text{Rank}(A) = 1$

nullity of $A = n - \text{Rank}(A)$

$= 3 - 1$

$= 2$

4(c) i. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 - 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \frac{x_2}{2}\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 2 & 0 & 4 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{r_2 = r_2 - 2r_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 4 & 4 \\ 0 & 1 & 1 \end{array}\right]$$

$$\downarrow \begin{array}{l} r_1 = r_1 + \frac{1}{2}r_2 \\ r_2 = \frac{1}{4}r_2 \\ r_3 = r_3 - \frac{1}{4}r_2 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

$\therefore \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$

$\therefore \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$ is in Nul(A)

ii. $\begin{bmatrix} -1 \\ 3 \end{bmatrix} \neq k\begin{bmatrix} 5 \\ -1 \end{bmatrix}$ where $k \in \mathbb{R}$

$\therefore \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ not in Col(A)

5a let $\det(A-\lambda I)=0$

$$\begin{vmatrix} 3-\lambda & -1 & -2 \\ 2 & -\lambda & -2 \\ 2 & -1 & -1-\lambda \end{vmatrix} = 0$$

$$-\lambda^3+2\lambda^2-\lambda=0$$

$$-\lambda(\lambda-1)^2=0$$

case $\lambda=0$:

$$\left[\begin{array}{ccc|c} 3 & -1 & -2 & 0 \\ 2 & 0 & -2 & 0 \\ 2 & -1 & -1 & 0 \end{array}\right] \xrightarrow[r_3=2r_1-3r_2]{r_2=r_2-r_3} \left[\begin{array}{ccc|c} 3 & -1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow[r_3=r_2-r_3]{r_1=\frac{1}{3}r_1} \left[\begin{array}{ccc|c} 1 & -1/3 & -2/3 & \\ 0 & 1 & -1 & \\ 0 & 0 & 0 & \end{array}\right]$$

$$\xrightarrow{r_1=r_1+\frac{1}{3}r_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore e_1$ is the eigenvector for $\lambda=0$

case $\lambda=1$:

$$\begin{bmatrix} 2 & -1 & -2 \\ 2 & -1 & -2 \\ 2 & -1 & -2 \end{bmatrix} \xrightarrow[r_3=r_3-r_2]{r_1=\frac{1}{2}r_1,\ r_2=r_2-r_3} \begin{bmatrix} 1 & -1/2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2+x_3 \\ x_2 \\ x_3 \end{bmatrix} = \frac{x_2}{2}\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore e_2$ is the eigenvector for $\lambda=1$

(b) $\therefore e_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

5b $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right] \xrightarrow[r_3=r_3-r_1]{r_2=r_2-r_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right]$

$\xrightarrow[r_3=r_2+r_3]{r_2=-r_3}$

$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xleftarrow[r_1=r_1-r_2+r_3]{r_3=-r_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$

$\therefore \{e_1, e_2, e_3\}$ is linearly independent

(c) From 5b, we know A is diagonalizable

$A = PBP^{-1}$ where $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{aligned} \therefore P^{-1}AP &= P^{-1}(PBP^{-1})P \\ &= (P^{-1}P)B(P^{-1}P) \\ &= B \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{aligned}$$